

Summary of assignment and bioinformatics tool analysis:

The Bioinformatics Toolbox				
Tool / Database	Pipeline Phase	Input Type	Primary Output	Core Purpose
NCBI BLAST	Phase 1	Single Sequence	Homology Matches	Sequence Similarity Search
Clustal Omega / MUSCLE	Phase 2	Multiple Sequences	Alignment Grid	Evolutionary Conservation
SWISS-MODEL / AlphaFold DB	Phase 3	Single Sequence	3D Coordinate Map	Structure Prediction
PyMOL / Chimera	Phase 3	3D Coordinate Map	3D Visual Rendering	Structure Visualization